

MorphoCLIP: Text-Supervised Contrastive Learning for Perturbation Matching in Cell Painting Images

Sukhrobbek Ilyosbekov
Northeastern University
Portland, ME, USA
ilyosbekov.s@northeastern.edu

Shubham Gajjar
Northeastern University
Portland, ME, USA
gajjar.shu@northeastern.edu

Rongfei Jin
Northeastern University
Portland, ME, USA
jin.ron@northeastern.edu

Abstract—Cell Painting microscopy produces multi-channel phenotypic images of cells under chemical or genetic perturbation, but matching those images back to the underlying perturbation remains a bottleneck for phenotypic drug discovery. We present MorphoCLIP, a contrastive learning framework that embeds Cell Painting images and natural-language descriptions of perturbations in a shared 512-dimensional space. The image branch passes each of five fluorescence channels through a frozen DINOv3 ViT-L/16 backbone and aggregates the resulting CLS tokens with a single-layer CrossChannelFormer; the text branch uses a frozen BioClinical ModernBERT with a trainable projection head. Training uses the Continuously Weighted Contrastive Loss (CWCL) to handle soft-positive relationships across perturbations that share a biological target, and the framework additionally supports cross-well alignment for batch correction. Only about 14M parameters are trained, against roughly 1.5B total in the closest prior text-based method. On a held-out CPJUMP1 validation split (2,220 wells drawn from 98 perturbations), text-to-image retrieval reaches Recall@10 of 24.3% with a median rank of 28, roughly $2.3\times$ a like-for-like random baseline of 10.4%; image-to-text retrieval is close to chance, which we analyze. Unlike prior contrastive methods restricted to small molecules, MorphoCLIP is trained jointly on compounds, CRISPR knockouts, and ORF overexpressions in a single model. We report these as preliminary results and set out the component ablations and perturbation-level evaluation needed to establish the contribution of each part of the design.

Index Terms—contrastive learning, cell painting, morphological profiling, CLIP, perturbation matching

I. INTRODUCTION

Phenotypic drug discovery aims to identify how a chemical or genetic perturbation changes the state of a cell, and to do so at a scale that target-specific assays cannot reach. This matters because the path from candidate to approved drug costs on the order of \$2B and takes about a decade [1], and roughly nine of ten compounds that enter clinical trials fail [2].

High-content microscopy offers a scalable alternative. The Cell Painting assay [3], [4] stains live cells with five fluorescent dyes covering DNA, endoplasmic reticulum, RNA/nucleoli, actin/Golgi, and mitochondria, producing a multi-channel image of cellular state. Morphological changes after perturbation encode the perturbation’s mechanism. The CPJUMP1 benchmark [5] casts this as a retrieval problem: given ~ 3 million field-of-view images covering 303 compounds, 160 CRISPR knockouts, and 176 ORF overexpres-

sions, can a model match perturbations that act on the same biological target?

Existing methods make uneven progress. The classical CellProfiler pipeline [6] extracts handcrafted shape, texture, and intensity features and recovers only 5–25% of expected compound–gene matches on CPJUMP1 [5]. Recent contrastive methods do better, but each covers a different part of the problem. CLOOME [7] pairs cell images with Morgan fingerprints and reaches $70\times$ random retrieval, but only for compounds. MolPhenix [8] improves on this with a pre-trained Phenom-1 encoder, again restricted to compounds. CellCLIP [9] replaces fingerprints with text prompts and a CrossChannelFormer over DINOv2-g (1.48B parameters), enabling all three perturbation types through language, but its cross-class mAP still trails CellProfiler. CWA-MSN [10] instead targets batch effects directly, using cross-well alignment for correction; with only 22M parameters it improves on CellCLIP by 9% on gene–gene retrieval, but it has no text branch and so cannot be queried in natural language.

We propose **MorphoCLIP**, a contrastive learning framework that brings text supervision, optional cross-well alignment for batch correction, and joint training over both small molecules and genetic perturbations into a single model. Following CLIP [11], MorphoCLIP aligns Cell Painting images and natural-language perturbation descriptions in a shared 512-dimensional space. Our working hypothesis is that text supervision and batch correction are complementary: language adds semantic structure that purely visual methods cannot recover, while cross-well alignment removes technical variation that otherwise swamps fine-grained biological signal. Whether the two combine as hoped is a question we frame here and leave to the planned ablations, since the results in this paper are from a text-only, batch-correction-off configuration.

The contributions of this paper are:

- 1) A parameter-efficient contrastive framework built on a frozen DINOv3 ViT-L/16 backbone [12] (~ 300 M frozen) and a CrossChannelFormer aggregator, with about 14M trainable parameters against CellCLIP’s 1.48B total.
- 2) A single model trained jointly on compounds, CRISPR knockouts, and ORF overexpressions through a shared text encoder, where prior contrastive methods are limited to compounds.

- 3) A design that supports optional cross-well alignment and a factorial ablation to attribute performance to text supervision, batch correction, and gene-inclusive training.
- 4) A preliminary CPJUMP1 evaluation reported against a like-for-like random baseline, in which text-to-image retrieval reaches Recall@10 of 24.3% ($\sim 2.3\times$ random) while image-to-text retrieval stays close to chance.

II. PROBLEM STATEMENT

Given a Cell Painting image $\mathbf{x} \in \mathbb{R}^{5 \times H \times W}$ (five fluorescence channels at 1080×1080 pixels) and a textual description t of the perturbation applied to the imaged cells, we learn encoders f_{img} and f_{txt} that map both modalities into a shared unit sphere:

$$f_{\text{img}}(\mathbf{x}), f_{\text{txt}}(t) \in \mathbb{R}^d, \quad \|f_{\text{img}}(\mathbf{x})\|_2 = \|f_{\text{txt}}(t)\|_2 = 1. \quad (1)$$

Matching image–text pairs should have high cosine similarity and non-matching pairs low similarity. We use $d = 512$.

The trained encoders support three retrieval tasks:

- **Text-to-image retrieval.** Given a perturbation description (e.g., “a compound that inhibits tubulin and disrupts cell division”), retrieve cell images that exhibit the corresponding phenotype.
- **Image-to-text retrieval.** Given a cell image, return the perturbation description that best matches the observed morphology.
- **Image-to-image retrieval.** Compare compound images against genetic perturbation images by cosine similarity to identify shared targets, the standard CPJUMP1 task. The text encoder shapes the embedding space during training but is not used at inference for this task.

Three properties of the data make the problem harder than standard vision–language alignment.

a) Batch effects.: The same perturbation produces visually different images across plates, days, and microscopes. Illumination drift, staining inconsistencies, and focus variation introduce technical variance that often dominates the biological signal. In CPJUMP1, replicability varies by modality (compounds replicate more cleanly than CRISPR) and by cell line (U2OS vs. A549) [5]. The mAP distribution is heavily right-skewed: most perturbations score close to zero.

b) Fuzzy ground truth.: Biological perturbations do not admit clean positive/negative labels in the way natural image–caption pairs do. Two compounds may share a primary target but diverge on secondary effects; a CRISPR knockout and a small molecule can converge on the same pathway through different mechanisms. Labels live on a continuum of similarity, which calls for soft-positive loss formulations rather than InfoNCE-style hard negatives.

c) Scale and compute.: CPJUMP1 contains 51 plates and 3.31 TiB of raw 16-bit TIFF images, covering ~ 3 million field-of-view images of ~ 75 million cells. End-to-end training of full vision and language encoders at this scale is not feasible on a single consumer GPU. We address this with a pre-extraction

and caching pipeline that decouples backbone forward passes from the contrastive training loop.

III. RELATED WORK

A. Morphological Profiling

Cell Painting [3] introduced the now-standard five-stain protocol for high-content morphological profiling and has become the dominant assay for high-throughput phenotypic screening [4]. The classical analysis tool, CellProfiler [6], extracts more than 1,000 handcrafted shape, texture, and intensity features per cell. On CPJUMP1 [5], this pipeline retrieves 5–25% of expected sister perturbations within a modality and 7–17% of CRISPR guide matches; cross-modal compound-to-gene matching is harder still.

Handcrafted features are bounded by the measurements their authors chose to compute and miss higher-order phenotypic patterns. Self-supervised vision transformers trained with DINO surpass CellProfiler on drug-target and gene-family classification with zero-shot transfer [13], which motivates the use of frozen foundation-model features for Cell Painting.

B. Contrastive Learning for Cell Painting

CLOOME [7] was the first CLIP-style framework for Cell Painting. It jointly trains a ResNet image encoder and a Morgan-fingerprint MLP and reaches top-1 retrieval $> 70\times$ random over $\sim 2,000$ candidates. CLOOME handles only chemical perturbations and treats the five channels as RGB, discarding channel-specific signal.

MolPhenix [8] replaces the trained image branch with a frozen Phenom-1 encoder, averages replicate embeddings, and introduces a soft-to-label (S2L) loss, gaining $8.78\times$ over CLOOME. It also covers compounds only and depends on a proprietary backbone.

CellCLIP [9] is the closest prior work. It contributes (i) templated text prompts (“A Cell Painting image of [cell_type] cells treated with [drug_name], SMILES: [SMILES_string]”), (ii) a CrossChannelFormer that processes each channel separately before cross-attention, (iii) attention-based MIL pooling across cells, and (iv) the Continuously Weighted Contrastive Loss (CWCL) for soft positives. CellCLIP runs on a DINOv2-g backbone (1.1B parameters, 1.48B total) and is evaluated on the Bray dataset [14] and on RxRx3-core gene–gene benchmarks (CORUM recall 0.354). Although CellCLIP supports all perturbation types through text, its cross-class mAP falls below the CellProfiler baseline, suggesting that text supervision on its own does not solve the batch-effect problem.

C. Batch-Effect Correction

CWA-MSN [10] addresses batch effects directly with cross-well alignment inside a masked siamese network. By aligning embeddings of cells subjected to the same perturbation across different wells and plates, it absorbs technical variance without an explicit batch label. With 22M parameters and 0.2M training images, CWA-MSN improves on CellCLIP by 9% and on OpenPhenom by 29% on gene–gene retrieval (CORUM recall 0.386). It has no text branch.

TABLE I

LANDSCAPE OF CELL PAINTING REPRESENTATION LEARNING METHODS. MORPHOCLIP IS DESIGNED TO COMBINE TEXT SUPERVISION, GENE-INCLUSIVE TRAINING, AND BATCH CORRECTION. THE PARAMETER COUNT IS TRAINABLE PARAMETERS FOR MORPHOCLIP AND TOTAL PARAMETERS FOR THE OTHER LEARNED METHODS.

Method	Image Encoder	Text	Pert. Types	Batch Corr.	Params
CellProfiler [6]	Handcrafted	No	All	Sphering	N/A
CLOOME [7]	ResNet	No	Compounds	No	~25M
MolPhenix [8]	Phenom-1	No	Compounds	No	~36M
CellCLIP [9]	DINOv2-g	Yes	All	No	1,477M
CWA-MSN [10]	ViT-S/16	No	Unimodal	CWA	22M
MorphoCLIP	DINOv3-L	Yes	All	CWA (opt.)	~14M

D. Foundation Models for Microscopy

DINOv2 [15] and DINOv3 [12] provide self-supervised visual representations that transfer well to biological imaging despite being trained on natural images [13]. DINOv3 adds gram anchoring and an improved training recipe over DINOv2. Channel Vision Transformers [16] show that per-channel processing followed by cross-channel attention is effective for multi-channel microscopy, which motivates our CrossChannelFormer aggregator.

On the text side, BioClinical ModernBERT [17] (150M parameters) is pre-trained on 53.5B tokens of PubMed, PMC, and clinical text, roughly $17\times$ the biomedical corpus used by PubMedBERT (110M, 3.1B tokens) in CellCLIP. Built on ModernBERT with RoPE, Flash Attention, and GeGLU activations, it can encode both small-molecule and gene descriptions in a single model, unlike SMILES-specialized encoders such as ChemBERTa.

E. Positioning of MorphoCLIP

Table I places MorphoCLIP relative to these methods. No prior method combines text supervision with both gene-inclusive training and batch-effect correction: CellCLIP has text but no gene-inclusive contrastive training and no batch correction; CWA-MSN has genes and batch correction but no text. MorphoCLIP is designed to fill that intersection with a parameter-efficient architecture (~14M trainable parameters against CellCLIP’s 1.48B total). The batch-correction component is optional and is evaluated in the planned ablation rather than in the results reported here.

IV. PROPOSED APPROACH

MorphoCLIP has two encoding branches and one contrastive objective. The image branch turns a five-channel Cell Painting field of view into a 512-dimensional unit vector; the text branch does the same for a natural-language prompt describing the perturbation. Figure 1 shows the full pipeline.

A. Image Encoder

Each Cell Painting site has five fluorescence channels (Table II). We process each channel independently: the single-channel image is replicated to pseudo-RGB and passed

TABLE II

CELL PAINTING FLUORESCENCE CHANNELS USED IN MORPHOCLIP.

Ch	Stain	Target
1	MitoTracker / Alexa 647	Mitochondria
2	Phalloidin / Alexa 568	Actin
3	WGA / Alexa 488	Golgi / plasma membrane
4	Concanavalin A / Alexa 488	Endoplasmic reticulum
5	Hoechst 33342	DNA / nucleus

through a frozen DINOv3 ViT-L/16 [12] (300M parameters), producing five 1024-dimensional CLS tokens per site,

$$\{\mathbf{z}_c\}_{c=1}^5, \quad \mathbf{z}_c \in \mathbb{R}^{1024}. \quad (2)$$

We use DINOv3 ViT-L rather than the DINOv2-g backbone of CellCLIP. The two have comparable feature quality on transfer benchmarks, but DINOv3 ViT-L is roughly $4\times$ smaller, which lets us pre-extract a full plate in about seven minutes on a single RTX 5080. Features are written to disk as .pt files (~3 GB per plate) and reused across every training run, removing the need to re-run the backbone each epoch.

a) *CrossChannelFormer (CCF)*.: To collapse the five channel tokens into a single site vector, we use a single-layer transformer encoder [18] with 4 attention heads, following the channel-aware processing of CellCLIP [9] and Channel ViT [16]. Each channel token is L2-normalized and augmented with a learnable channel-type embedding $\mathbf{e}_c$ so the transformer can tell the five stains apart. A learnable aggregation token $\mathbf{q}$ is prepended, and the transformer output at that position is taken as the aggregated site representation:

$$\mathbf{h}_{\text{site}} = [\text{CCF}(\mathbf{q}, \mathbf{z}_1 + \mathbf{e}_1, \dots, \mathbf{z}_5 + \mathbf{e}_5)]_0. \quad (3)$$

This lets the model attend across channels and read out cross-channel co-occurrences, for example a joint signal in the mitochondrial and ER channels. We also report a baseline that drops the CCF and averages the five L2-normalized channel tokens directly.

b) *Site-to-well aggregation*.: The unit of training is a well. Per-site representations are combined into one well vector by masked mean pooling over the sites in the well, which keeps the well vector well-defined regardless of how many sites were imaged.

c) *Image projection head*.: A two-layer MLP with LayerNorm, GELU, and dropout ($p = 0.3$) maps the pooled well vector $\mathbf{h}_{\text{img}}$ into the shared 512-dimensional space:

$$f_{\text{img}}(\mathbf{x}) = \text{L2Norm}(\mathbf{W}_2 \cdot \text{Drop}(\text{GELU}(\text{LN}(\mathbf{W}_1 \cdot \mathbf{h}_{\text{img}})))), \quad (4)$$

with $\mathbf{W}_1 \in \mathbb{R}^{512 \times 1024}$ and $\mathbf{W}_2 \in \mathbb{R}^{512 \times 512}$.

B. Text Encoder

We convert the perturbation metadata into a verbose prompt using one of four modality-specific templates, populated from the CPJUMP1 metadata and external biological databases. Each template names the cell line, the perturbation, and its available annotations:

MorphoCLIP Architecture

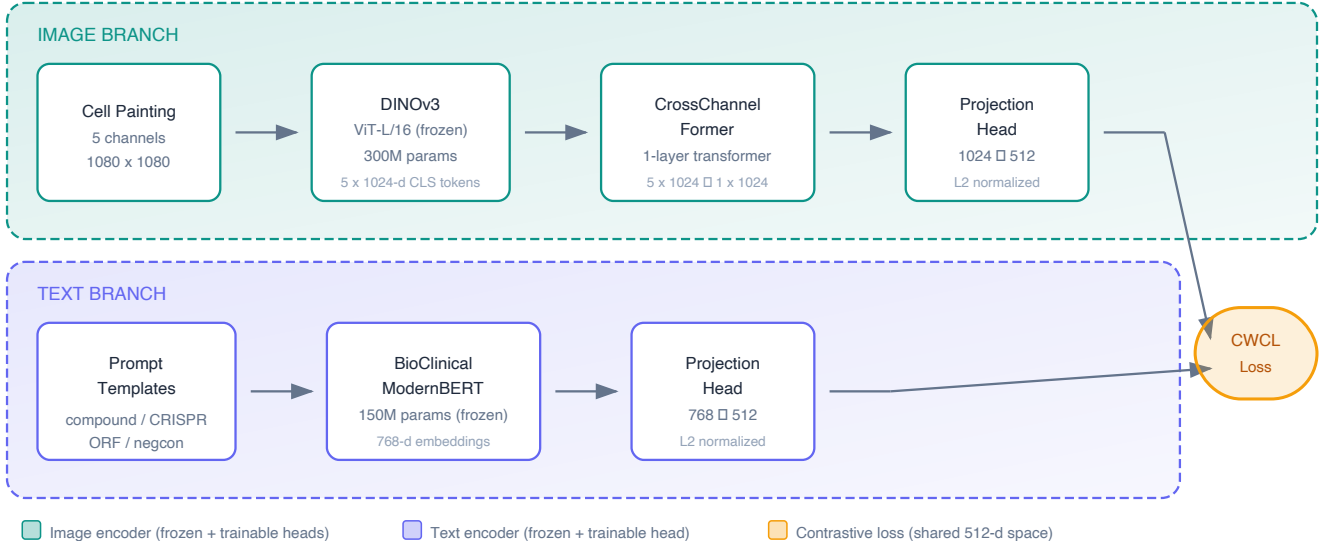


Fig. 1. MorphoCLIP architecture. A five-channel Cell Painting image (1080×1080) is passed channel-by-channel through a frozen DINOv3 ViT-L/16, producing five 1024-d CLS tokens. A single-layer CrossChannelFormer (4 heads) attends over the channel tokens together with a learnable aggregation token, whose output becomes the site representation; site representations are mean-pooled into a well vector and projected to the shared 512-d space. The text prompt is encoded by frozen BioClinical ModernBERT (150M parameters) and projected to the same space. CWCL pulls matching pairs together and pushes non-matches apart. Only the shaded components (CrossChannelFormer and the two projection heads) are trained, about 14M parameters.

- **Compound:** “Cell Painting morphological profile of $\{cell_line\}$ cells treated with the compound $\{compound_name\}$. Chemical structure (SMILES): $\{smiles\}$. Known target gene: $\{target_gene\}$. Target protein function: $\{gene_function\}$. Perturbation modality: chemical compound.”
- **CRISPR:** “... with CRISPR-Cas9 knockout of gene $\{gene_symbol\}$. Gene description: $\{gene_description\}$. Gene function: $\{gene_function\}$. Perturbation modality: CRISPR knockout.”
- **ORF:** “... overexpressing gene $\{gene_symbol\}$ via open reading frame construct. Gene description: $\{gene_description\}$. Gene function: $\{gene_function\}$. Perturbation modality: ORF overexpression.”
- **Negative control:** “... treated with DMSO vehicle control. No active perturbation applied.”

Missing annotation fields are filled with “unknown”. Compound annotations come from ChEMBL and the CPJUMP1 metadata; gene descriptions and functions come from UniProt.

We encode each prompt with a frozen BioClinical ModernBERT [17] (150M parameters, 768-d CLS token), pre-trained on 53.5B tokens of PubMed, PMC, and clinical text. A trainable projection head matches the image head and maps the CLS token to the 512-d space:

$$f_{\text{txt}}(t) = \text{L2Norm}(\mathbf{W}_4 \cdot \text{Drop}(\text{GELU}(\text{LN}(\mathbf{W}_3 \cdot \mathbf{h}_{\text{txt}})))), \quad (5)$$

with $\mathbf{W}_3 \in \mathbb{R}^{512 \times 768}$ and $\mathbf{W}_4 \in \mathbb{R}^{512 \times 512}$. The 768-d BERT features are pre-computed and cached separately from the

projected 512-d outputs, which removes the BERT forward pass from the training loop and gives roughly an order-of-magnitude speedup per epoch.

C. Training Objective

a) *Continuously Weighted Contrastive Loss (CWCL).*: Standard InfoNCE [19] treats every off-diagonal pair as a hard negative, which is the wrong assumption when two compounds share a target. Following CellCLIP [9] and CWA-MSN [10], we use CWCL with continuous similarity weights: same broad_sample $\rightarrow 1.0$, same target gene but different compound $\rightarrow 0.5\text{--}0.8$, otherwise $\rightarrow 0.0$. For a batch of N image-text pairs with logits $\mathbf{S} = \tau \cdot \mathbf{F}_{\text{img}} \mathbf{F}_{\text{txt}}^\top$ and weight matrix $\mathbf{W} \in [0, 1]^{N \times N}$, the per-pair term is

$$\ell_i^{\text{img} \rightarrow \text{txt}} = - \sum_{j=1}^N \tilde{w}_{ij} \log \frac{\exp(S_{ij})}{\sum_{k=1}^N \exp(S_{ik})}, \quad (6)$$

where $\tilde{w}_{ij} = w_{ij} / \sum_k w_{ik}$ normalizes weights row-wise. The full loss is the symmetric average,

$$\mathcal{L}_{\text{CWCL}} = \frac{1}{2N} \sum_{i=1}^N (\ell_i^{\text{img} \rightarrow \text{txt}} + \ell_i^{\text{txt} \rightarrow \text{img}}). \quad (7)$$

The temperature τ is a learnable parameter (LogitScale module), initialized to $\ln(1/0.07)$ and clamped at $\ln(100)$.

b) *Cross-Well Alignment (CWA).*: Batch effects are a dominant confound in Cell Painting. As an optional component, MorphoCLIP supports cross-well alignment: image

embeddings are grouped by plate within each batch, the per-plate mean is subtracted, and the result is re-normalized to the unit sphere,

$$\tilde{\mathbf{f}}_w = \frac{\mathbf{f}_w - \bar{\mathbf{f}}_{p(w)}}{\|\mathbf{f}_w - \bar{\mathbf{f}}_{p(w)}\|_2}, \quad \bar{\mathbf{f}}_p = \frac{1}{|W_p|} \sum_{w \in W_p} \mathbf{f}_w, \quad (8)$$

where $p(w)$ is the plate of well w and W_p is the set of wells from plate p in the current batch. CWA is applied to image embeddings only and requires a batch sampler that draws from multiple plates. It is disabled in the runs reported here and is isolated in the planned ablation (Section V-G).

D. Optimization

We train with AdamW [20] at learning rate 10^{-4} , weight decay 0.2, $(\beta_1, \beta_2) = (0.9, 0.999)$, and $\varepsilon = 10^{-8}$. The schedule is a 100-step linear warmup followed by cosine decay over 100 epochs, with the reported checkpoint selected by validation loss (early best, epoch 9). We use FP16 mixed precision and gradient clipping at norm 1.0. The batch size is 256 wells, which sets the pool of in-batch negatives for the contrastive objective. The trainable parameter count is about 14M (the CrossChannelFormer plus the two projection heads), so the optimizer state fits comfortably in the 16 GB of an RTX 5080. With pre-extracted features, an epoch takes 3–5 minutes wall-clock.

V. EXPERIMENTAL SETUP

A. Dataset

We evaluate on the CPJUMP1 pilot dataset [5], a public benchmark from the JUMP Cell Painting Consortium hosted on the Cell Painting Gallery (AWS S3, prefix `cpg0000-jump-pilot`). CPJUMP1 contains 51 plates and 3.31 TiB of raw 16-bit TIFF images. The collection covers three perturbation types (303 small-molecule compounds, 160 CRISPR gene knockouts, 176 ORF overexpressions), two cell lines (U2OS and A549), and several timepoints. Each gene target is linked to at least two compounds, which allows systematic evaluation of compound–gene matching. Each plate has 384 wells with 7–16 sites per well; each site produces five fluorescence channels at 1080×1080 pixels plus three brightfield channels that we do not use.

B. Data Pipeline

We use a stream-extract-delete pipeline to keep disk usage bounded. Raw TIFFs (~ 58 – 118 GiB per plate) are downloaded to scratch space, passed through the frozen DINOv3 backbone, and removed once the per-channel CLS features are written. This compresses 3.31 TiB of raw data into roughly 153 GB of cached features (~ 3 GB per plate). Extraction takes about seven minutes per plate on an RTX 5080, or roughly six hours for all 51 plates. Text embeddings are pre-computed in the same way: each perturbation prompt is encoded by frozen BioClinical ModernBERT and the 768-d CLS tokens are cached to disk.

The unit of training is a well with all of its sites aggregated. The cached feature tensor for a well has shape $(S, 5, 1024)$,

where S is the number of sites. After the CrossChannelFormer produces one vector per site, the site vectors are mean-pooled into a single well representation.

C. Train/Validation/Test Split

The runs reported here use a perturbation-stratified split. Each perturbation (`broad_sample`) is hashed deterministically into train, validation, or test at an 80/10/10 ratio, with all three modalities present in every split and all wells of a perturbation kept together, so no perturbation is split across partitions. We also implement two alternative splits for later use: a gene-level holdout, in which no target gene appears in both train and test, and a CellCLIP-style within-slice split that matches CellCLIP’s evaluation protocol.

D. Evaluation Protocol

We report numbers on the held-out validation split, which contains 2,220 wells drawn from 98 perturbations (42 compound, 34 CRISPR, 22 ORF), so each perturbation is represented by about 24 replicate wells. Retrieval ranks each query against all 2,220 wells, and a query is counted correct when any well sharing its perturbation appears in the top k . Because replicate wells share an identical text embedding, both retrieval directions face the same positive structure. Both directions are reported:

- **Recall@ k** for $k \in \{1, 5, 10\}$: the fraction of queries whose first correct match appears in the top- k retrieved candidates.
- **Median rank**: the median position of the first correct match in the full ranked list (lower is better).

A like-for-like random baseline, computed by ranking the 2,220 wells uniformly at random against the same positive structure, gives Recall@1/5/10 of approximately 1.1%/5.3%/10.4% and a median rank near 64. Because each perturbation has many replicate positives, this baseline is much higher than the naïve single-positive value; we use it as the reference throughout. This well-level protocol differs from the standard CPJUMP1 setup, which aggregates replicates into one profile per perturbation and ranks over the perturbation set; moving to that protocol is the first item of future work (Section VII).

E. Baselines

We compare against three baselines. Each was reported on a different dataset or task formulation, so the comparisons are indicative rather than strictly controlled, and we flag the relevant differences in the results section.

- **CellProfiler** [6]. Handcrafted features on CPJUMP1 sister-perturbation matching, fraction retrieved 4.3–25.1% (mean $\sim 13\%$) [5].
- **CellCLIP** [9]. Contrastive model on the Bray dataset [14] with text-to-profile retrieval. Best Recall@10 7.4%; CORUM gene–gene recall 0.354.
- **CWA-MSN** [10]. Self-supervised siamese network with cross-well alignment. CORUM gene–gene recall 0.386 with 22M parameters.

TABLE III
FACTORIAL ABLATION DESIGN.

Variant	CWA	Genes	Text	Tests
Full MorphoCLIP	✓	✓	✓	Complete method
No text	✓	✓		CWA-MSN-like
No genes	✓		✓	CellCLIP + CWA
No CWA		✓	✓	Batch corr. effect

F. Implementation Details

Batch size 256 wells. AdamW optimizer with $\text{lr} = 10^{-4}$, weight decay 0.2, $(\beta_1, \beta_2) = (0.9, 0.999)$, $\varepsilon = 10^{-8}$. Schedule: 100-step linear warmup followed by cosine decay over 100 epochs, with the reported checkpoint selected by validation loss. FP16 mixed precision; gradient clipping at norm 1.0. CrossChannelFormer: 1 transformer layer, 4 attention heads. Projection heads: hidden width 512, dropout 0.3. Temperature initialized to $\ln(1/0.07)$ and clamped at $\ln(100)$. The headline number uses seed 42; the planned ablation runs use three seeds (42, 1337, 2024) for variance estimates.

G. Planned Ablations

We design a 2^3 factorial ablation over three binary factors (text supervision, CWA batch correction, and gene-inclusive training) to attribute performance to each component:

A second planned ablation varies prompt richness across four levels: (1) perturbation name only, (2) name plus target and function, (3) name plus target plus SMILES or gene description, and (4) full description with pathway annotations. The goal is to measure how much of the gain comes from injecting structured domain knowledge into the prompt, and in particular whether SMILES strings carry useful signal through BioClinical ModernBERT, which was not pre-trained on chemical notation.

VI. RESULTS AND DISCUSSION

We stress at the outset that the numbers below are preliminary. They come from a single text-only, batch-correction-off configuration (seed 42), evaluated at the well level rather than the perturbation-aggregated level of the standard CPJUMP1 protocol. We report them against a like-for-like random baseline and are explicit about where the model does and does not beat chance.

A. Cross-Modal Retrieval Performance

Table IV reports retrieval performance of MorphoCLIP on the CPJUMP1 validation split alongside the published CellCLIP and CellProfiler numbers and the random baseline for our evaluation. The baselines were measured on different datasets and task formulations, so the cross-method comparisons are indicative; we discuss the caveats below.

The strongest result is text-to-image Recall@10 of 24.3%, about $2.3\times$ the like-for-like random baseline of 10.4% on the same 2,220-well pool, with a median rank of 28 against a random median near 64. Recall@5 (14.9%) is likewise about

TABLE IV
CROSS-MODAL RETRIEVAL. MORPHOCLIP IS EVALUATED ON THE CPJUMP1 VALIDATION SPLIT (2,220 WELLS, 98 PERTURBATIONS, ~ 24 REPLICATE WELLS EACH). THE RANDOM ROW IS A LIKE-FOR-LIKE BASELINE FOR THIS WELL-LEVEL POOL. CELLCLIP NUMBERS ARE FROM THE ORIGINAL PAPER ON THE BRAY DATASET; CELLPROFILER USES PROFILE-BASED SISTER-PERTURBATION MATCHING ON CPJUMP1.

Model	Direction	R@1	R@5	R@10
CellProfiler	sister matching	FR: 4.3–25.1%		
CellCLIP	perturb $\rightarrow$ profile	1.18	4.49	7.37
	profile $\rightarrow$ perturb	1.25	4.82	7.39
Random (our val)	either direction	1.09	5.33	10.37
MorphoCLIP	image $\rightarrow$ text	5.72	6.71	7.07
	text $\rightarrow$ image	3.51	14.86	24.32

$2.8\times$ random. Text-to-image retrieval is therefore modestly but consistently above chance.

Image-to-text retrieval is close to chance in aggregate. Recall@1 (5.7%) is several times the random rate of 1.1%, so the model does resolve a subset of exact matches, but Recall@10 (7.1%) sits below the random 10.4%, and the median rank of 289 is far worse than the random median of 64. In this direction the model surfaces a few confident matches but does not order the broader positive set better than chance. We read this as a ceiling of the current well-level setup and single configuration, not as a settled property of the method.

B. Retrieval Asymmetry

The two directions share the same positive structure: the mask of same-perturbation pairs is symmetric, so the random baseline is identical for both. The asymmetry in the learned model therefore comes from the embeddings, not from the candidate pool. Two factors plausibly contribute. First, the text side has only 98 distinct vectors (one per perturbation) with strong semantic structure from BioClinical ModernBERT, which makes a text query an easier anchor to match images against. Second, the image embeddings, aggregated from frozen DINOv3 features under heavy batch variation, are noisier, so an image query resolves its matching text less reliably. Aggregating replicate wells into a single per-perturbation profile before retrieval would remove much of this noise and is the natural next evaluation.

C. Comparison with Baselines

a) *Versus CellCLIP.*: The comparison is not strictly controlled: CellCLIP is evaluated on the Bray dataset over aggregated profiles, whereas MorphoCLIP is evaluated on CPJUMP1 over individual wells with a different candidate pool. On the more directly comparable image-to-text direction, MorphoCLIP’s Recall@10 of 7.07% is close to CellCLIP’s profile-to-perturb Recall@10 of 7.39% – but both sit at or below our random baseline for this well-level setup, so this should be read as a rough parity of order rather than evidence of a strong effect. The contribution we can defend without further experiments is efficiency and coverage: about 14M trainable parameters against CellCLIP’s 1.48B total, with joint

TABLE V

WALL-CLOCK PROFILE OF THE TRAINING LOOP ON A SINGLE RTX 5080 (16 GB VRAM). THE LoRA ROW IS AN INDICATIVE ESTIMATE FROM A PARTIAL RUN, INCLUDED TO MOTIVATE THE PRE-EXTRACTION DESIGN RATHER THAN AS A MEASURED BASELINE.

Configuration	Train. Params	VRAM	Time/epoch
Pre-extracted + CCF (ours)	~14M	~2 GB	3–5 mi
Frozen DINOv3 + LoRA ($r=8$, est.)	~20M	~14 GB	45–90 mi

training over all three perturbation types instead of compounds only.

b) Versus CellProfiler.: On CPJUMP1, CellProfiler’s fraction retrieved ranges from 4.3% to 25.1% (mean ~13%) depending on modality and matching task [5]. MorphoCLIP’s text-to-image Recall@10 sits inside that range, on the easier text-query direction and under a different protocol, so the numbers are not directly comparable. The qualitative difference, independent of any single number, is that MorphoCLIP supports natural-language querying of an image database, which CellProfiler does not.

c) Versus CWA-MSN.: A head-to-head comparison requires running both methods on the same gene–gene retrieval benchmarks (CORUM, HuMAP, Reactome). CWA-MSN’s CORUM recall of 0.386 with 22M parameters is the relevant target. The factorial ablation in Table III is designed to answer the narrower question of whether text supervision adds anything over cross-well alignment alone.

D. Feature Analysis

We inspect the pre-extracted DINOv3 features to check that channel structure survives the frozen backbone. On plate BR00116991 (500 sites), a variance decomposition attributes 31.1% of feature variance to differences between channels, 37.1% to differences between sites, and 31.8% to residual variation (Figure 2). The between-channel component is large enough to suggest that the per-channel encoding carries channel-specific signal, even though DINOv3 was trained on natural images. A PCA over the 5×1024 -dimensional feature space reaches 90% of variance with 23 principal components (Figure 4), indicating moderate intrinsic dimensionality.

Cross-site same-channel similarity (mean 0.84) is higher than cross-channel similarity at the same site (mean 0.75), a gap of 0.09 that holds for every channel pair (Figure 3). The DNA/nucleus channel is the most distinct, with a mean similarity of 0.73 to the other four channels, consistent with its very different biological content.

E. Computational Efficiency

Pre-extraction is the main reason MorphoCLIP is cheap to iterate on (Table V). The DINOv3 forward pass is the dominant cost; once features are cached, the contrastive training loop is small enough to fit comfortably on a single 16GB GPU.

Feature extraction is a one-time cost of about seven minutes per plate (~six hours for all 51). After that, an epoch over

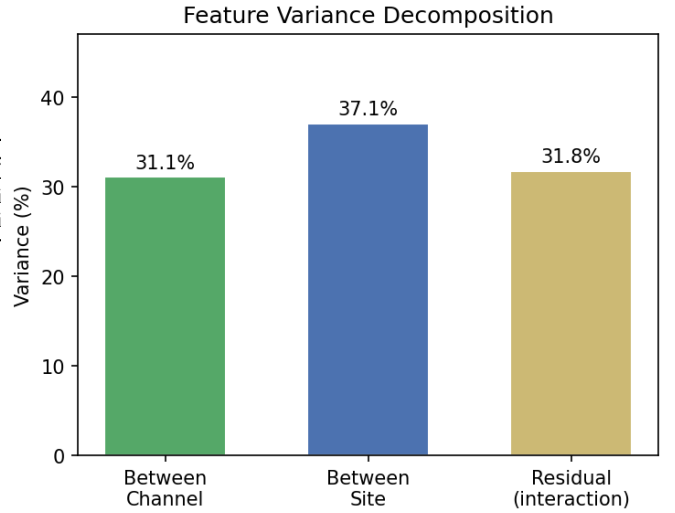


Fig. 2. Variance decomposition of the cached DINOv3 features on plate BR00116991: between-channel, between-site, and residual components.

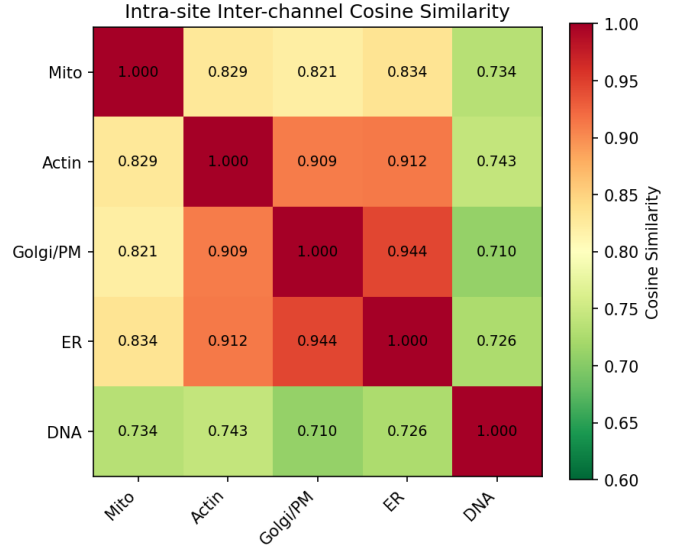


Fig. 3. Mean cosine similarity between fluorescence channels in the cached feature space. The DNA/nucleus channel is the most distinct.

the cached features takes 3–5 minutes wall-clock, which is what makes it practical to run the planned ablation across many configurations on a single GPU. Both DINOv3 (300M parameters) and BioClinical ModernBERT (150M parameters) stay frozen throughout; only the CrossChannelFormer and the two projection heads are updated.

VII. CONCLUSION

We have presented MorphoCLIP, a contrastive learning framework that maps Cell Painting images and natural-language perturbation descriptions into a shared 512-dimensional space. The design brings together three capabili-

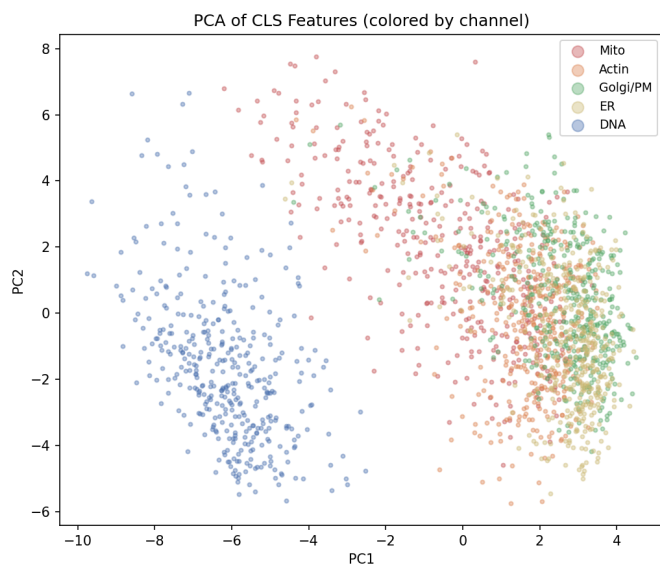


Fig. 4. Cumulative explained variance of a PCA over the 5×1024 -d feature space; 90% is reached at 23 components.

ties that prior work has only paired in twos – text supervision, optional cross-well batch alignment, and joint training over both small molecules and genetic perturbations – and trains on a single 16GB GPU thanks to a pre-extraction pipeline that caches the frozen-backbone outputs. About 14M parameters are trained, against 1.48B total in the closest text-based baseline.

The empirical results are preliminary and mixed. On the CPJUMP1 validation split, text-to-image retrieval reaches Recall@10 of 24.3% with a median rank of 28, roughly $2.3\times$ a like-for-like random baseline; image-to-text retrieval, by contrast, is close to chance in aggregate. These numbers come from a single seed, a text-only and batch-correction-off configuration, and a well-level evaluation protocol, so they bound what the current setup shows rather than establish the method.

If a perturbation-level evaluation and the planned ablations confirm that the signal holds up, the use cases this design points toward include searching for existing drugs whose cellular signature matches a target phenotype, suggesting a likely mechanism of action from an unknown compound’s neighbors in embedding space, and querying image databases in plain English without engineering bespoke features.

A. Limitations and Future Work

The most immediate limitation is the evaluation protocol. We rank individual wells, where each perturbation carries about 24 replicate positives; the standard CPJUMP1 protocol instead aggregates replicates into one profile per perturbation and ranks over the perturbation set. Re-running the evaluation at the perturbation level is the first priority, both to compare cleanly against prior work and because aggregation should

reduce the image-side noise that leaves image-to-text retrieval near chance today.

The second limitation is that the design’s central claim is asserted rather than measured. Cross-well alignment is disabled in the reported runs, and the factorial ablation in Table III that would attribute performance to text supervision, batch correction, and gene-inclusive training has not yet been run. Until it lands – across the three seeds and the held-out test split – the contribution of each component is a hypothesis. The prompt-richness ablation will separately test whether SMILES strings and gene-function text add signal through BioClinical ModernBERT or are inert.

Beyond closing those gaps, three directions are worth pursuing. First, scale: CPJUMP1 is the pilot, and the full JUMP corpus contains $\sim 140,000$ perturbations across many labs, microscopes, and cell lines, which is the right setting to test generalization. Second, richer prompts: pulling pathway and mechanism annotations from UniProt, Reactome, and DrugBank should strengthen the text encoder’s biological grounding. Third, zero-shot transfer: running the trained encoders on Cell Painting datasets from labs and cell lines not seen during training would show whether the learned representations are general or specific to CPJUMP1.

We plan to release model weights, the evaluation suite, and an interactive natural-language retrieval demo on publication.

REFERENCES

- [1] O. J. Wouters, M. McKee, and J. Luyten, “Estimated research and development investment needed to bring a new medicine to market, 2009–2018,” *JAMA*, vol. 323, no. 9, pp. 844–853, 2020.
- [2] D. Sun, W. Gao, H. Hu, and S. Zhou, “Why 90% of clinical drug development fails and how to improve it?” *Acta Pharmaceutica Sinica B*, vol. 12, no. 7, pp. 3049–3062, 2022.
- [3] M.-A. Bray, S. Singh, H. Long, and A. E. Carpenter, “Cell painting, a high-content image-based assay for morphological profiling using multiplexed fluorescent dyes,” *Nature Protocols*, vol. 11, no. 9, pp. 1757–1774, 2016.
- [4] S. Seal *et al.*, “Cell painting: A decade of discovery and innovation in cellular imaging,” *Nature Methods*, 2024.
- [5] S. N. Chandrasekaran, J. Ackerman, E. Alix, D. M. Ando, J. Arevalo, M. Berber, N. Beresford, L. Bornholdt, B. Bostock *et al.*, “JUMP cell painting dataset: morphological impact of 136,000 chemical and genetic perturbations,” *Nature Methods*, vol. 21, pp. 1114–1121, 2024.
- [6] A. E. Carpenter, T. R. Jones, M. R. Lamprecht, C. Clarke, I. H. Kang, O. Friman, D. A. Guertin, J. H. Chang, R. A. Lindquist, J. Moffat *et al.*, “CellProfiler: image analysis software for identifying and quantifying cell phenotypes,” *Genome Biology*, vol. 7, no. 10, p. R100, 2006.
- [7] A. Sánchez-Fernández, T. Langer, G. Ecker *et al.*, “CLOOME: Contrastive learning unlocks bioimaging databases for queries with chemical structures,” *Nature Communications*, 2023.
- [8] P. Fradkin *et al.*, “How molecules impact cells: Unlocking contrastive PhenoMolecular retrieval,” in *Advances in Neural Information Processing Systems*, 2024.
- [9] M. Lu, E. Weinberger, C. Kim, and S.-I. Lee, “CellCLIP: Learning perturbation effects in cell painting via text-guided contrastive learning,” in *Advances in Neural Information Processing Systems*, 2025.
- [10] P.-J. Huang, Y.-H. Liao, S. Kim, N. Park, J. Park, and D. Shin, “Efficient cell painting image representation learning via cross-well aligned masked siamese network,” *arXiv preprint arXiv:2509.19896*, 2025.
- [11] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark *et al.*, “Learning transferable visual models from natural language supervision,” in *International Conference on Machine Learning*. PMLR, 2021, pp. 8748–8763.

- [12] O. Simeoni *et al.*, “DINOv3: Self-supervised learning via gram anchoring,” *arXiv preprint arXiv:2508.10104*, 2025.
- [13] N. Moshkov *et al.*, “Self-supervision advances morphological profiling by unlocking powerful image representations,” *Scientific Reports*, 2025.
- [14] V. Ljosa, K. L. Sokolnicki, and A. E. Carpenter, “Annotated high-throughput microscopy image sets for validation,” *Nature Methods*, vol. 9, no. 7, pp. 637–637, 2012.
- [15] M. Oquab, T. Darcet, T. Moutakanni, H. Vo, M. Szafraniec, V. Khalidov, P. Fernandez, D. Haziza, F. Massa, A. El-Nouby *et al.*, “DINOv2: Learning robust visual features without supervision,” *Transactions on Machine Learning Research*, 2024.
- [16] Y. Bao, S. Sivanandan, and T. Karaletsos, “Channel vision transformers: An image is worth $1 \times 16 \times 16$ words,” in *International Conference on Learning Representations*, 2024.
- [17] T. Sounack *et al.*, “BioClinical ModernBERT: A state-of-the-art long-context encoder for biomedical and clinical NLP,” *arXiv preprint arXiv:2506.10896*, 2025.
- [18] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, “Attention is all you need,” *Advances in Neural Information Processing Systems*, vol. 30, 2017.
- [19] A. van den Oord, Y. Li, and O. Vinyals, “Representation learning with contrastive predictive coding,” in *Advances in Neural Information Processing Systems*, 2018.
- [20] I. Loshchilov and F. Hutter, “Decoupled weight decay regularization,” in *International Conference on Learning Representations*, 2019.